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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

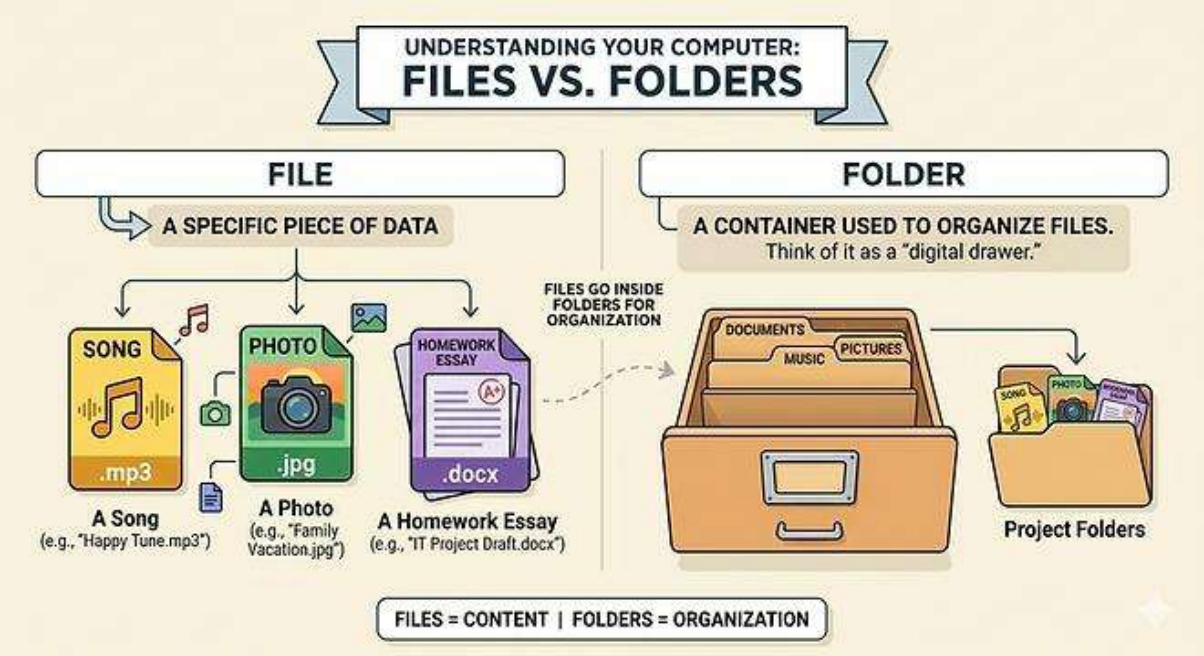
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 36: File Handling in Python



Learning Objectives

By the end of this lesson, students will be able to:

- Understand what a **file** is.
- Explain why computers use files.
- Open and close files using Python.
- Write information into a file.
- Read information from a file.
- Add new information to an existing file.
- Understand common file modes.
- Create a simple **Student Notes File** project.

1. What is a File?

A **file** is a place where a computer stores information permanently.

For example, when you save:

- School notes
- Photos
- Music
- Videos
- Documents
- Marks

the computer stores them as **files**.

Simple Example

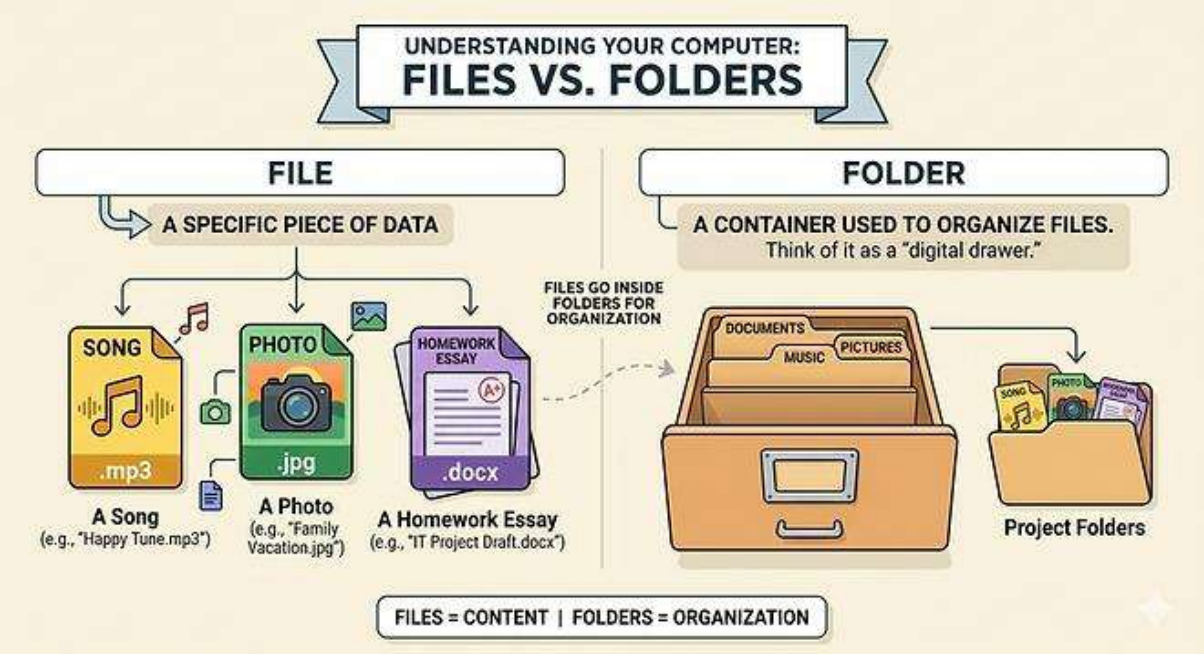
Think about your school notebook.

Notebook → contains information

Similarly:

Computer File → contains information

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2. What is a Folder?

A **folder** is a container used to organize files.

Imagine a school bag:

School Bag

→ Math Notebook
→ Science Notebook
→ English Notebook

Similarly:

Computer Folder

→ notes.txt
→ marks.txt
→ homework.txt

Remember

Folder = Container
File = Information stored inside the container

3. Why Do We Need File Handling?

Normally, information stored inside a Python program can disappear when the program ends.

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Example:

name = "Aisha"

print(name)

The program remembers name only while it is running.

If we want to **save information for later**, we can use a file.

Example

A school program can save:

Student Name: Aisha

Class: 8

Marks: 92

into a file.

4. What is File Handling?

File Handling means using a computer program to:

1. Create a file
2. Open a file
3. Read a file
4. Write to a file
5. Add information
6. Close a file

Python makes file handling easy.

5. The open() Function

Python uses the open() function to open a file.

Syntax

open("filename", "mode")

Example:

file = open("notes.txt", "r")

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Here:

- notes.txt → File name
- "r" → Reading mode
- file → Variable that represents the opened file

6. File Modes

A **mode** tells Python what we want to do with the file.

Mode	Meaning
R	Read
W	Write
A	Append
X	Create a new file

Easy Memory Trick

R → Read 📖

W → Write 🖍️

A → Add ➕

X → Create 🆕

7. Reading a File

Suppose we have a file called:

school.txt

It contains:

Welcome to Python!

Python is fun.

We can read it using:

file = open("school.txt", "r")

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```
data = file.read()

print(data)

file.close()
```

Output

```
Welcome to Python!
Python is fun.
```

8. Understanding read()

The read() method reads the contents of the file.

```
data = file.read()
```

Think of it like opening a book and reading all the pages.

File → read() → Python

9. Writing to a File

We can write information into a file using "w" mode.

```
file = open("student.txt", "w")

file.write("My name is Aisha.")

file.close()
```

Python creates student.txt if it doesn't already exist.

The file contains:

```
My name is Aisha.
```

Important: w Mode

Be careful!

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If the file already contains information, "w" mode can **replace the existing content**.

Example:

Before:

Hello

Welcome to Python

Then:

```
file = open("student.txt", "w")

file.write("New Message")

file.close()
```

After:

New Message

The old content is replaced.

10. Append Mode

Sometimes we don't want to delete old information.

We want to **add new information**.

Use:

"a"

Example:

```
file = open("student.txt", "a")


file.write("\nI am learning Python.")


file.close()
```

Now the file contains:

My name is Aisha.

I am learning Python.

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11. Create a New File

Python can create a new file using "x" mode.

file = open("newfile.txt", "x")

file.close()

This creates:

newfile.txt

If the file already exists, Python will give an error.

12. Closing a File

After working with a file, we should close it.

file.close()

Why?

Because closing a file:

- Saves changes properly
- Releases computer resources
- Prevents unwanted problems

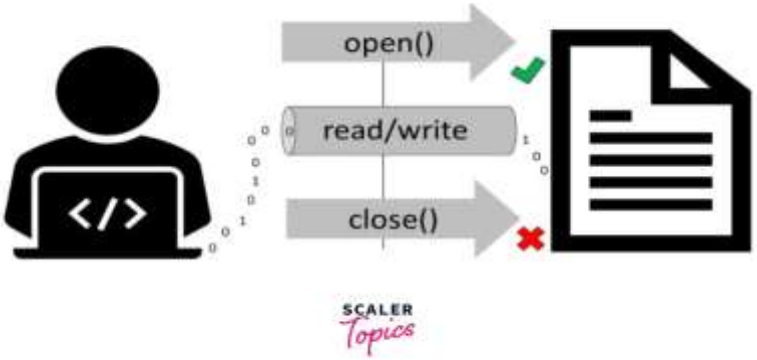
Easy Example

Open the door → use the room → close the door

Similarly:

Open file → work with file → close file

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SCALER Topics

13. readline()

readline() reads **one line** from a file.

Example:

```
file = open("school.txt", "r")

line = file.readline()

print(line)

file.close()
```

If the file contains:

Python is easy.

Python is powerful.

Python is fun.

The output may be:

Python is easy.

14. readlines()

readlines() reads all lines and gives them as a list.

```
file = open("school.txt", "r")
```

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```
lines = file.readlines()


print(lines)


file.close()
```

Example output:

```
['Python is easy.\n', 'Python is powerful.\n', 'Python is fun.\n']
```

15. Reading Using a Loop

We can read a file line by line.

```
file = open("school.txt", "r")


for line in file:
    print(line)


file.close()
```

This is useful when a file has many lines.

16. The with Statement

Python provides an easier and safer way to work with files.

```
with open("school.txt", "r") as file:
    data = file.read()
    print(data)
```

Python automatically closes the file when the with block finishes.

Student-Friendly Idea

Instead of:

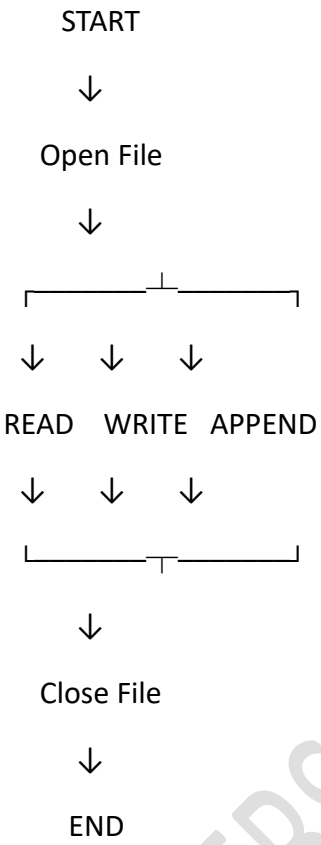
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Open → Work → Remember to Close

with does:

Open → Work → Automatically Close

17. File Handling Flow



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Practical Session

Practical Example 1 – Create a Notes File

```
with open("notes.txt", "w") as file:
    file.write("Python is easy to learn.")
    file.write("\nPython is fun.")
```

File Content

Python is easy to learn.

Python is fun.

Practical Example 2 – Read Notes

```
with open("notes.txt", "r") as file:
    notes = file.read()

print(notes)
```

Output

```
Python is easy to learn.
Python is fun.
```

Practical Example 3 – Add Notes

```
with open("notes.txt", "a") as file:
    file.write("\nI love programming!")
```

The file now contains:

```
Python is easy to learn.
Python is fun.
I love programming!
```

Practical Example 4 – Student Information

```
name = input("Enter your name: ")
age = input("Enter your age: ")
school = input("Enter your school: ")

with open("student.txt", "w") as file:
    file.write("Student Information\n")
    file.write("Name: " + name + "\n")
```

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```
    file.write("Age: " + age + "\n")
    file.write("School: " + school + "\n")

print("Information saved!")
```

Example Input

Enter your name: Aisha

Enter your age: 13

Enter your school: ABC School

File Content

Student Information

Name: Aisha

Age: 13

School: ABC School

Practical Example 5 – Student Marks

```
name = input("Enter student name: ")
marks = input("Enter marks: ")

with open("marks.txt", "w") as file:
    file.write("Student: " + name + "\n")
    file.write("Marks: " + marks)

print("Marks saved successfully!")
```

23. Mini Project – My Shopping List

Students can create a shopping list file.

```
with open("shopping.txt", "w") as file:
    file.write("Milk\n")
```

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```
    file.write("Bread\n")
    file.write("Apples\n")
    file.write("Pencils\n")

print("Shopping list created!")
```

Now read it:

```
with open("shopping.txt", "r") as file:
    print(file.read())
```

Output

Milk

Bread

Apples

Pencils

24. Classroom Activity – My Favorite Things

Ask students to create:

favorite.txt

and save:

My Favorite Things

Favorite Color: Blue

Favorite Food: Pizza

Favorite Sport: Cricket

Favorite Subject: Computer

Favorite Animal: Dog

Challenge

Ask students to read the file and display it on the screen.

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25. Activity – Complete the Code

Fill in the blanks:

file = open("hello.txt", "___")

file.___("Hello Python!")

file.___()

Answers

w

write

close

26. Quick Comparison

Task	Method
Open file	open()
Read everything	read()
Read one line	readline()
Read all lines	readlines()
Write	write()
Close	close()

27. Common Mistakes

Mistake 1: Forgetting quotation marks

❌

open(notes.txt, "r")

✅

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```
open("notes.txt", "r")
```

Mistake 2: Forgetting to close

```
file.close()
```

Or better:

```
with open("notes.txt", "r") as file:
    print(file.read())
```

Mistake 3: Using the wrong mode

If you want to add information:

❌

```
open("notes.txt", "w")
```

✅

```
open("notes.txt", "a")
```

28. Real-Life Applications

File handling is used in many places:

Schools → Student records

Hospitals → Patient information

Banks → Account records

Shops → Product information

Games → Save game data

Apps → User information

29. Practical Lab Task

Create a "My Profile" File

Ask students to enter:

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- Name
- Age
- Class
- School
- Favorite subject

Save the information in:

myprofile.txt

Expected Program

```
name = input("Name: ")
age = input("Age: ")
student_class = input("Class: ")
school = input("School: ")
subject = input("Favorite Subject: ")

with open("myprofile.txt", "w") as file:
    file.write("MY PROFILE\n")
    file.write("Name: " + name + "\n")
    file.write("Age: " + age + "\n")
    file.write("Class: " + student_class + "\n")
    file.write("School: " + school + "\n")
    file.write("Favorite Subject: " + subject + "\n")

print("Profile saved successfully!")
```

Challenge Activity

Create a program that:

1. Creates todo.txt
2. Stores 5 tasks

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3. Reads the tasks
4. Adds one new task
5. Displays all tasks

Example:

My To-Do List

1. Complete homework
2. Read Python notes
3. Practice coding
4. Play outside
5. Prepare school bag

Class Quiz

1. What is a file?

A) A computer game
B) A place to store information
C) A keyboard
D) A programming language

Answer: B

2. Which function opens a file?

A) start()

B) open()

C) file()

D) begin()

Answer: B

3. Which mode is used for reading?

A) r

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B) w

C) a

D) x

Answer: A

4. Which mode is used for writing?

A) r

B) w

C) a

D) read

Answer: B

5. Which mode adds information without replacing existing content?

A) r

B) w

C) a

D) x

Answer: C

6. Which method reads all file content?

A) read()

B) write()

C) append()

D) close()

Answer: A

7. Which method writes information?

A) read()

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B) write()

C) open()

D) get()

Answer: B

8. Why do we close a file?

A) To delete it

B) To finish working with it properly

C) To rename it

D) To print it

Answer: B

Homework

Task 1

Create a file called:

school.txt

Store:

- School name
- Student name
- Class
- Section
- Favorite subject

Task 2

Read the file and display its contents.

Task 3

Add:

I love learning Python!

using append mode.

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Task 4

Create a friends.txt file and store the names of **5 friends**.

Learning Check

Students should now be able to answer:

What is a file?
What is file handling?
What is open()?
What is "r" mode?
What is "w" mode?
What is "a" mode?
How do we read a file?
How do we write to a file?
Why do we use close()?
Why is with open() useful?

One-Minute Revision

FILE HANDLING

```
|
├── open()
|
├── Read
|   ├── read()
|   ├── readline()
|   └── readlines()
|
├── Write
|   └── write()
|
├── Append
```

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```
|
└── Close
    └── close()
```

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